

Gated Causal Alignment: consulting a neighbor only when its evidence would change the action

A value-of-alignment gate for multi-specialist robot control, evaluated in a synthetic causal model and a physics drawer task

Summary. A robot arm, a mobile base and a verifier each hold evidence about a different part of a drawer task's causal graph. Before each decision the arm may align with a neighbor, at a declared cost. We propose a gate that opens an alignment edge only when the expected decision gain exceeds that cost, the single-agent observation-refinement criterion applied to agent-to-agent edges. In exact enumeration over a synthetic causal model the gate stays within 0.017 utility per decision of always-align while paying 16% less, and its cost stays flat as redundant specialists are added. In 72 held-out physics episodes it matches always-align on compliant completion (67%) and utility, preserves 99.4% of its decisions, and sends 27% fewer messages at three specialists and 57% fewer at five. The fusion rule matters more than the gate: a fixed-weight diffusion pool loses 43% of utility even with every edge open. No controller violated a constraint on held-out configurations; isolation cost completion and time instead.

DECISIONS PRESERVED

99.4%

gate vs always-align

MESSAGES SAVED

27%

N=3; 57% at N=5

COMPLIANT COMPLETION

67%

equal to always-align; ceiling given infeasible cases

CAN-POOL UTILITY LOSS

43%

fixed-weight diffusion vs exact fusion

1. Problem and idea

Process-compliant manipulation can fail while a proxy succeeds: a robot that pulls a drawer handle makes progress on handle displacement whether the drawer opens or the whole cabinet slides. Specialists hold complementary causal evidence (the base sees the cabinet move; the verifier knows the force limit), but continuous alignment is costly and fixed-weight consensus can wash out a confident neighbor. The question is which neighbors to consult, and when.

Existing lines of work answer two neighbouring questions: when one agent should pay to refine its own observation, and how agents should combine beliefs once they share. This work sits between them.

$$\text{VOA}_{ij}(t) = E[U(a \star (b_i \oplus b_j))] - E[U(a \star (b_i))] - c_{ij}$$

The arm holds beliefs p (the cabinet will slide) and q (the next pull crosses the force limit) and three skills: pull, reposition (brace), release (back off). A neighbor's report is a binary signal with calibrated true-positive and true-negative rates. The gate computes the preposterior value of each closed edge, opens the best positive one, updates, and repeats. Its precondition is the refinement theorem's: an edge has strict value only when the arm's belief aliases cases with different optimal actions and the report separates them. A finite check over 4,332 cases agreed in 4,326; the 6 others are exact ties.

2. Experimental design

Design space. An instance is $(F, T, N, G, \{G_i\}, \Omega, c, \Pi_{\text{fuse}}, \Pi_{\text{gate}}, \Phi)$: task family and horizon, number of specialists, task graph and subgraphs, overlap, cost, fusion rule, gate policy, verifier spec. Outcomes: compliant completion (drawer open, zero violations), violations, unnecessary stops, decision preservation against always-align, regret against the oracle gate, messages per decision.

Stage A: synthetic POSCM, exact

Stage B: physics drawer task

Force drives handle displacement, then the drawer or the cabinet moves; a force limit is exogenous. The arm's own evidence has a declared accuracy; the base and verifier report with accuracies 0.95 and 0.99; edge cost 0.1. Worlds, own evidence and all neighbor reports are enumerated, so every number is an expectation.

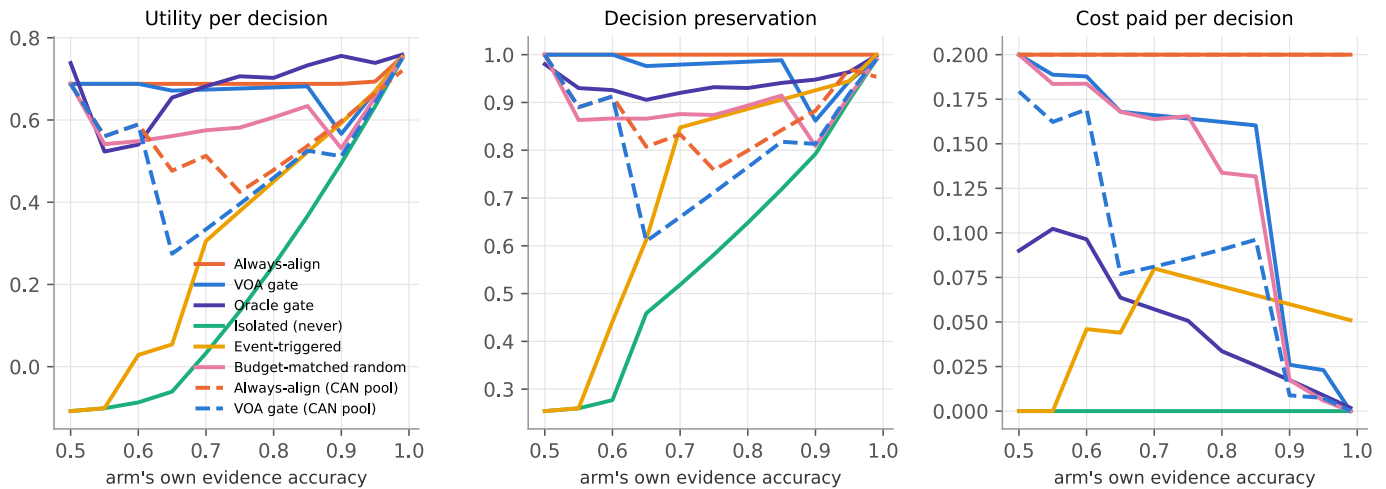
Sweeps: own-evidence quality (overlap), cost, prior, and N up to 12 with redundant and irrelevant specialists.

A MuJoCo scene: cabinet on a slide joint (anchored or free), drawer with rail resistance, a heavy base whose plate braces the cabinet, a pull modelled as an external force pair, a stiff-start handle variant. Specialists fit class-balanced logistic models on 35,409 steps from 72 training configurations (limits vary so only the verifier knows the limit). Held-out: 24 unseen configurations $\times$ 3 seeds. Report rates calibrated on training data: base 1.00/0.83, verifier 0.95/0.94; the arm alone reaches 0.88 on sliding.

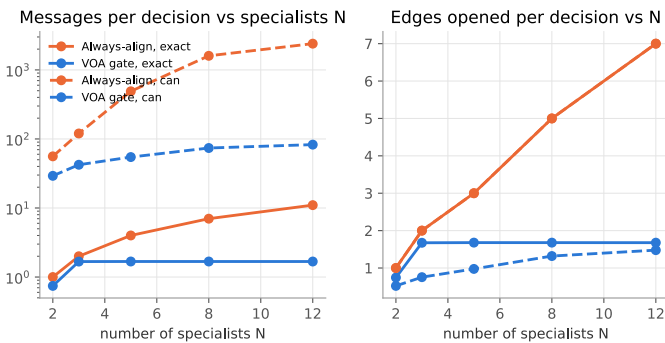
Controllers. Centralized network (one classifier over all features, no messages); isolated arm; always-align with exact fusion or a fixed-weight CAN diffusion pool; event-triggered; budget-matched random (the gate's pilot open rates); the VOA gate; an oracle gate that opens an edge only if the true latent would change the action (a reference, since it still acts on noisy reports).

3. Results

Stage A



Utility, decision preservation and cost paid against the arm's own evidence accuracy. At 0.65 the gate pays 16% less than always-align for a utility gap of 0.017; at 0.95 it pays 88% less. Dashed: the fixed-weight CAN pool, which loses decisions even with every edge open.



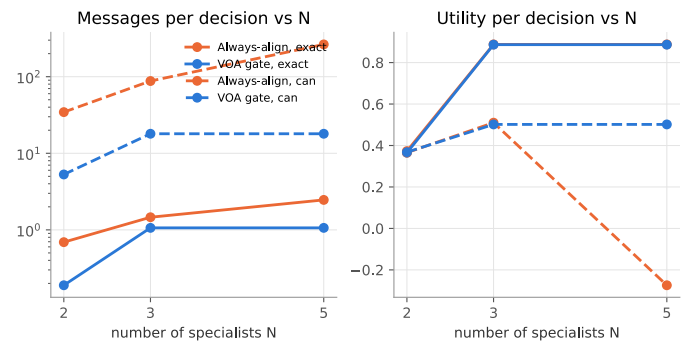
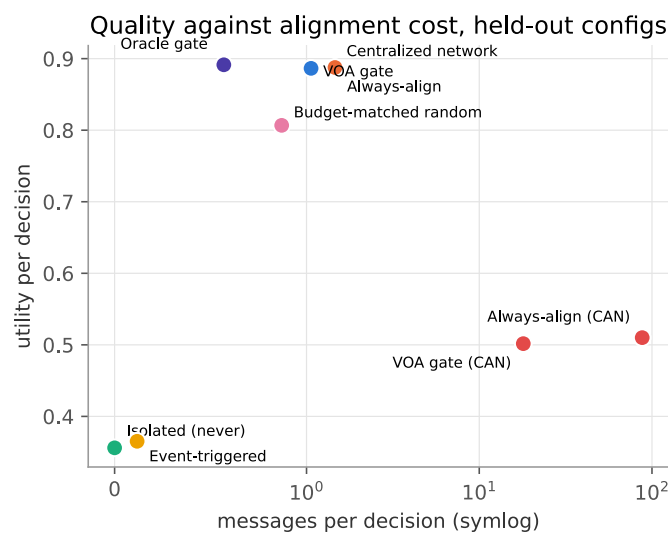
Adding redundant and irrelevant specialists: the gate opens 1.68 edges per decision at $N=12$ versus 1.68 at $N=3$; always-align grows to 11 messages (exact) or 2,400 (ring diffusion to consensus).

Milestone. With the arm's own evidence saying "fine" ($p = 0.19$, $q = 0.12$) the arm alone would pull. The base edge is worth 0.47 and the verifier edge 0.31, both above the cost of 0.1. A sliding report flips the action to reposition; a near-limit report to release. Complementary evidence changes the compliant action, which was the proposal's first deliverable.

Raising the declared cost closes edges smoothly. The budget-matched random control pays the same as the gate at every cost level and loses utility, because it opens the wrong edges.

Stage B: held-out physics episodes (72 per controller)

CONTROLLER	COMPLIANT	VIOLATIONS	STOPS / DEC.	PRESERVED	UTILITY / DEC.	STEPS	MSGS / DEC.
Centralized network	67%	0.00	0.064	99.7%	0.887	188	1.45
Isolated (never)	33%	0.00	0.482	55.2%	0.356	286	0.00
Always-align, exact	67%	0.00	0.063	100.0%	0.888	189	1.46
Event-triggered	33%	0.00	0.475	55.9%	0.365	286	0.12
Budget-matched random	67%	0.00	0.134	90.3%	0.807	260	0.87
VOA gate, exact	67%	0.00	0.065	99.4%	0.886	192	1.06
Oracle gate (ref.)	67%	0.00	0.066	98.2%	0.891	229	0.57
Always-align, CAN pool	67%	0.00	0.361	67.9%	0.510	247	87.80
VOA gate, CAN pool	67%	0.00	0.368	67.2%	0.502	247	17.99



From N=3 to N=5 always-align's cost grows 68% while the gate's does not move; the always-align CAN pool collapses at N=5 (utility -0.27).

Quality against cost. The gate sits beside always-align and the centralized network on utility at 27% fewer messages; CAN-pool variants are far right and below.

N	ALWAYS EXACT	GATE EXACT	ALWAYS CAN	GATE CAN	UTIL. EXACT A/G	UTIL. CAN A/G
2	0.69	0.19	34	5.3	0.37 / 0.37	0.37 / 0.37
3	1.46	1.06	88	18.0	0.89 / 0.89	0.51 / 0.50
5	2.46	1.06	264	18.0	0.89 / 0.89	-0.27 / 0.50
FAMILY	T=120		T=300		T=600	
anchored	-0.06 / 19%		+0.00 / 27%		+0.00 / 31%	
free cabinet	-0.06 / 19%		+0.00 / 27%		+0.00 / 31%	

Δ compliant completion (gate – always) / message saving by horizon.

PERTURBATION	ALWAYS	GATE	EVENT	ALWAYS MSGS	GATE MSGS
Reports delayed 1 step	67%	67%	33%	1.47	1.05
Reports delayed 3 steps	67%	67%	33%	1.48	1.05
Base dropped mid-episode	67%	67%	33%	0.65	0.41
Base map corrupted (30%)	28%	42%	0%	1.77	0.87

Compliant completion and messages per decision under delayed reports, a dropped base and a corrupted base map. Nominal: always 67%, gate 67%, event 33%.

4. Interpretation

- **The gate keeps the decision and drops the traffic.** It matched always-align on compliant completion, utility (0.886 vs 0.888) and stops, preserved 99.4% of decisions and sent 27% fewer messages. The oracle reference (0.57 messages per decision) shows headroom: the gate is conservative because the arm's own evidence is weak, so the base edge is usually worth its cost.
- **Cost scales with decision-relevant edges, not agents.** The gate never consulted the redundant head camera or the irrelevant torso; its messages stayed at 1.06 per decision from N=3 to N=5 while always-align rose to 2.46. Stage A shows the same up to twelve specialists.
- **The fusion rule matters more than the gate.** The fixed-weight CAN pool cut utility by 43% and preservation to 68% with every edge open, cost 88 messages per decision, and collapsed at N=5. This reproduces the earlier grid-pursuit finding inside a robot task: the gate should sit in front of an exact or confidence-weighted fusion rule.
- **Isolation cost completion and time, not violations.** No controller violated a constraint on held-out configurations; the arm's own learned model braces early. Isolation showed up as stalls (33% completion) and stops (0.48 per decision): an arm that releases without the verifier's limit backs off to the prior and never recovers.
- **A centralized network is as good on quality.** One classifier over all features matched always-align on every quality metric. The gate's advantage is cost and robustness to a corrupted neighbor (42% vs 28% compliant completion), not decision quality.
- **Limits.** Report accuracies are state-independent calibrations; the task has two binary latents; the horizon grid shows no family dependence because both cabinet types were braced early; the process-safety advantage did not materialise because no controller violated.

5. Next steps and research direction

- **State-dependent report accuracies** conditioned on the neighbor's own evidence strength, so the gate sees when a neighbor currently knows nothing; expected to close part of the gap to the oracle.
- **Confidence-weighted and convergence-matched fusion** behind the gate, so the fusion axis is a fair contest rather than a fixed-weight pool.
- **Learned subgraphs.** Discover each specialist's subgraph online and feed the discovered overlap to the gate; the cost claim predicts cost tracks the discovered number of decision-relevant edges.
- **Harder process constraints** the arm cannot anticipate (anchoring that changes mid-episode, a contact-dependent limit), so the safety value of alignment is measured rather than only its efficiency.
- **Shared harness.** Package the Task (environment, subgraphs, verifier spec, cost model) and Solver (gate, fusion) interfaces so the same gate runs on grid-pursuit and real-time-strategy environments and on a full mobile-manipulation simulator.
- **Sequential gate** that values an edge by its effect on later beliefs and probes.

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